College Algebra System of Three Linear Equations

1) Solve the system of equations

$x + y + z = 12$

$x - y + z = 30$

$2x + 3y + z = 8$

a) Write the matrix form of the above system of equations.

b) Row Reduced Echelon Form (RREF) of matrix = ?

c) solution to system of equations: x =____ y =_____ z =_____

2) Solve the system of equations

$-2x + 3y + 4z = 10$

$x + 2y + 3z = 30$

$5x + 4y + z = 13$

a) Write the matrix form of the above system of equations.

b) Row Reduced Echelon Form (RREF) of matrix = ?

c) solution to system of equations: x =____ y =_____ z =_____

3) Solve the system of equations

$4x - 0.5y + 0.4z = 3$

$11x + 2y + 5z = 11$

$2x + 3y + 7z = 12$

a) Write the matrix form of the above system of equations.

b) Row Reduced Echelon Form (RREF) of matrix = ?

c) solution to system of equations: x =____ y =_____ z =_____

4) Solve the system of equations

$x + z = 12$

$-y + z = 30$

$12x + 13y + z = 8$

a) Write the matrix form of the above system of equations.

b) Row Reduced Echelon Form (RREF) of matrix = ?

c) solution to system of equations: x =____ y =_____ z =_____

5) Solve the system of equations

$x + y = 1$

$-x + 2z = 12$

$2x + 3y + z = 7$

a) Write the matrix form of the above system of equations.

b) Row Reduced Echelon Form (RREF) of matrix = ?

c) solution to system of equations: x =____ y =_____ z =_____

6) Solve the system of equations

$y + 2z = 8$

$-3y + 3z = 10$

$2x + 4y + 5z = 9$

a) Write the matrix form of the above system of equations.

b) Row Reduced Echelon Form (RREF) of matrix = ?

c) solution to system of equations: x =____ y =_____ z =_____

7) At a movie theater, there is one price for students, one for adults, and another for senior citizens.
Family A bought 2 student tickets and 2 adult tickets for $40.
Family B bought 3 student tickets, 2 adult ticket, and 2 senior citizen tickets for $58.
Family C bought 2 of each for $50.

a) Write a system of equations that can be used to find the cost of each ticket

b) Solve the system to find the cost of each ticket.